Utsav Sharma

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EDUCATION

Indian Institute of Technology Roorkee 2023 – 2027
Integrated B.S.–M.S., Mathematics and Computing CGPA: 8.05 / 10
Class XII (CBSE): 94.0% · Class X (CBSE): 95.4%

TECHNICAL SKILLS

Languages: Go, Rust, Python, TypeScript, C++, SQL, Solidity, ZK DSLs
Backend & Systems: PostgreSQL, Temporal, REST APIs, OAuth 2.0, Webhooks, Idempotency, Docker, GCP, GitHub Actions
Cryptography: Zero-knowledge proofs, zkVMs, R1CS, Poseidon2, WHIR, EIP-7702
Data & ML: NumPy, pandas, scikit-learn, LightGBM, XGBoost, SHAP, Survival analysis, Reinforcement learning

EXPERIENCE

Software Engineer | Trinity (AI project manager for software teams) Jul 2026 – Oct 2026
- Built the shared integration layer in **Go** behind GitHub, Linear, Slack, Granola and Notion, and the Notion integration itself.
- Built a common **meeting-notes layer** in PostgreSQL for every note-taking tool, cut over in three releases, and the Meetings backend on it: calendar feed, post-meeting task suggestions and briefs.
- Shipped task and milestone **completion forecasts** from a **competing-risks** survival model fit by maximum likelihood on censored task histories, refit nightly per project; beat the default in a staging backtest (Brier 0.114 vs 0.248).
- Shipped the **pre-standup brief**, a personal Slack summary on Temporal that reads and messages only its owner.

Protocol Engineer | Atheon Jan 2026 – Jun 2026
- Built zero-knowledge **passport age verification** in Noir for World Foundation's ProveKit as a chain of 4 to 5 circuits linked by Poseidon2 commitments, roughly halving peak proving memory (1.84 GB to 982 MB).
- Fixed two **passport-forgery vulnerabilities** from a security audit (unbound RSA exponent, smuggled signing key); upgraded Noir from beta.11 to beta.19.
- Built a prototype **Jolt zkVM** prover on the hash-based **WHIR** commitment over the Goldilocks field that proves real RISC-V programs, and the design for moving Jolt off Dory.

Ethereum Protocol Fellow | Ethereum Foundation, Cohort 6 Jun 2025 – Nov 2025
- Built a **Rust** benchmarking framework for Ethereum consensus state transitions across 5 zkVMs (run time, cycles, proving cost), including the Jolt, Pico and Zisk backends.
- Prototyped a block-proving service in the **Ream** consensus client that gossips proofs to peers.

Software Engineer Intern | Enclave Jul 2025 – Nov 2025
- Built **MagicSpend++**'s Solana **Anchor** vault with signed, nonce-checked borrows, its Circle Gateway borrow path, one-transaction **EIP-7702** borrow-and-spend and USDC or SOL settlement, on 4 EVM mainnets and Solana devnet.
- Integrated it with **Privy and Turnkey** wallets and with **Fireplace**, a Polymarket trading app, for bets funded from Solana.

PROJECTS

Transaction Fraud Detection and Risk Scoring System | Finance Club May 2026 – Jun 2026
- Built a fraud-scoring model with **LightGBM and XGBoost** that routes each payment to **approve, review, or block**, tuning the two cut-offs to send fewer payments to manual review than a **rules-based baseline** without losing recall.
- Kept features **leakage-safe** by building transaction history only from earlier events, and validated on **time-ordered splits** so the model is always tested on payments that come after its training data.
- Served the model through a **FastAPI** endpoint that returns each decision with the **SHAP** attributions behind it.

Reinforcement Learning-based Market Making Agent using Deep Q Networks | Finance Club May 2025 – Jun 2025
- Built **Gym-style market-making environments** on minute-level limit-order-book data, one with buy/sell/hold actions and one that shifts both quotes around the mid, and trained **PPO and Dueling DQN** agents on them in TensorForce.
- Shaped the reward as damped PnL plus spread capture minus a **quadratic inventory penalty**; the damping counts every loss but only part of each gain, which steers the agent toward earning the spread rather than taking directional bets.
- Simulated fills from each minute's **buy/sell order-flow imbalance** and evaluated on held-out trading days, fitting the feature scaler on training days only so **no look-ahead** leaks into the state.

ACHIEVEMENTS

- **Winner**, ETHOnline 2024 and Agentic Ethereum 2025 (ETHGlobal); **Finalist**, Based India Buildathon 2024.
- Recipient, **INSPIRE Scholarship for Higher Education**, Government of India, for academic performance.

POSITIONS OF RESPONSIBILITY

- **Blockchain Society IITR** (Feb 2024 – Present): organised the StackTooDeep and Srishti developer events; ran technical sessions.
- **Finance Club IITR** (Feb 2024 – Present): organised Finvista and Quantathon; ran workshops on probability and algorithmic trading.